

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems

COURSE CODE: CSA0504

COURSE OUTCOME COVERED

CO1: Analyze DBMS architecture, different data models, schema types, and design

ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and

relationships. (BL1, BL2)

Activity Tool : Technical Presentation (Low)

Assessment Tool Weightage: 10%

QUESTIONS		
Q. No.	Question	Bloom's Level
1	Create and present an ER diagram for an Online Library System, explaining each component clearly.	BL2 – Understand
2	Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.	BL2 – Understand
3	Present the concept of data independence (logical and physical) and its significance in DBMS architecture.	BL2 – Understand
4	Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.	BL1 – Remember
5	Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.	BL2 – Understand

Code of Conduct:

I R.PRANATHI(192525076) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Pranathi

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism ; minimal contribution or ethical awareness



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1-AT3-TECHNICAL PRESENTATION

CSA0504 : Database Management Systems

PRESENTED BY :

R. Pranathi(192525076)

Question 1: ER Diagram for an Online Library System

Introduction

What is an ER Diagram?

- ER stands for Entity Relationship Diagram.
- It is used to design databases.
- It shows entities, attributes, and relationships.
- It helps organize data efficiently.

ER Diagram Components

Entities

Student

Book

Librarian

Issue

Attributes

Student ID

Book ID

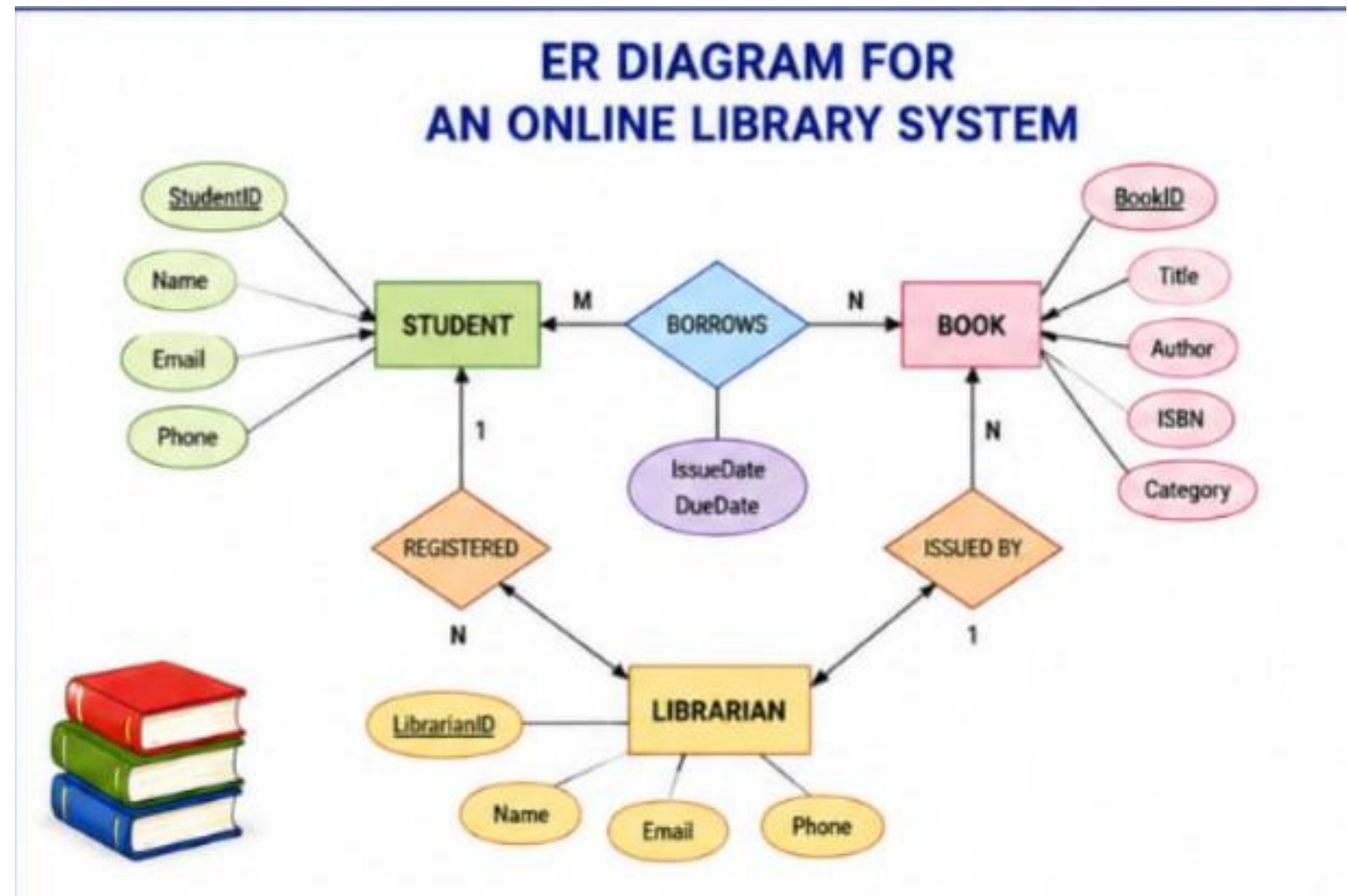
Book Title

Issue Date

Relationships

Student borrows Book.

Librarian issues Book.





Advantages & Conclusion

Advantages:

- Easy database design.
- Reduces data redundancy.
- Improves data organization.
- Easy maintenance.

Conclusion:

ER diagrams are the foundation of database design.



Question 2: EER Model – University Case Study

Introduction

- **Enhanced Entity Relationship (EER) Model** is an advanced version of ER (Entity-Relationship) model used for designing complex databases.
- It supports **inheritance**, allowing subclasses to inherit attributes from a superclass.
- **Generalization** combines similar entities into one higher-level entity.
- **Specialization** divides a general entity into specific sub-entities.
- It helps represent real-world systems more accurately and efficiently.

University Case Study

Superclass

- Person

Subclasses

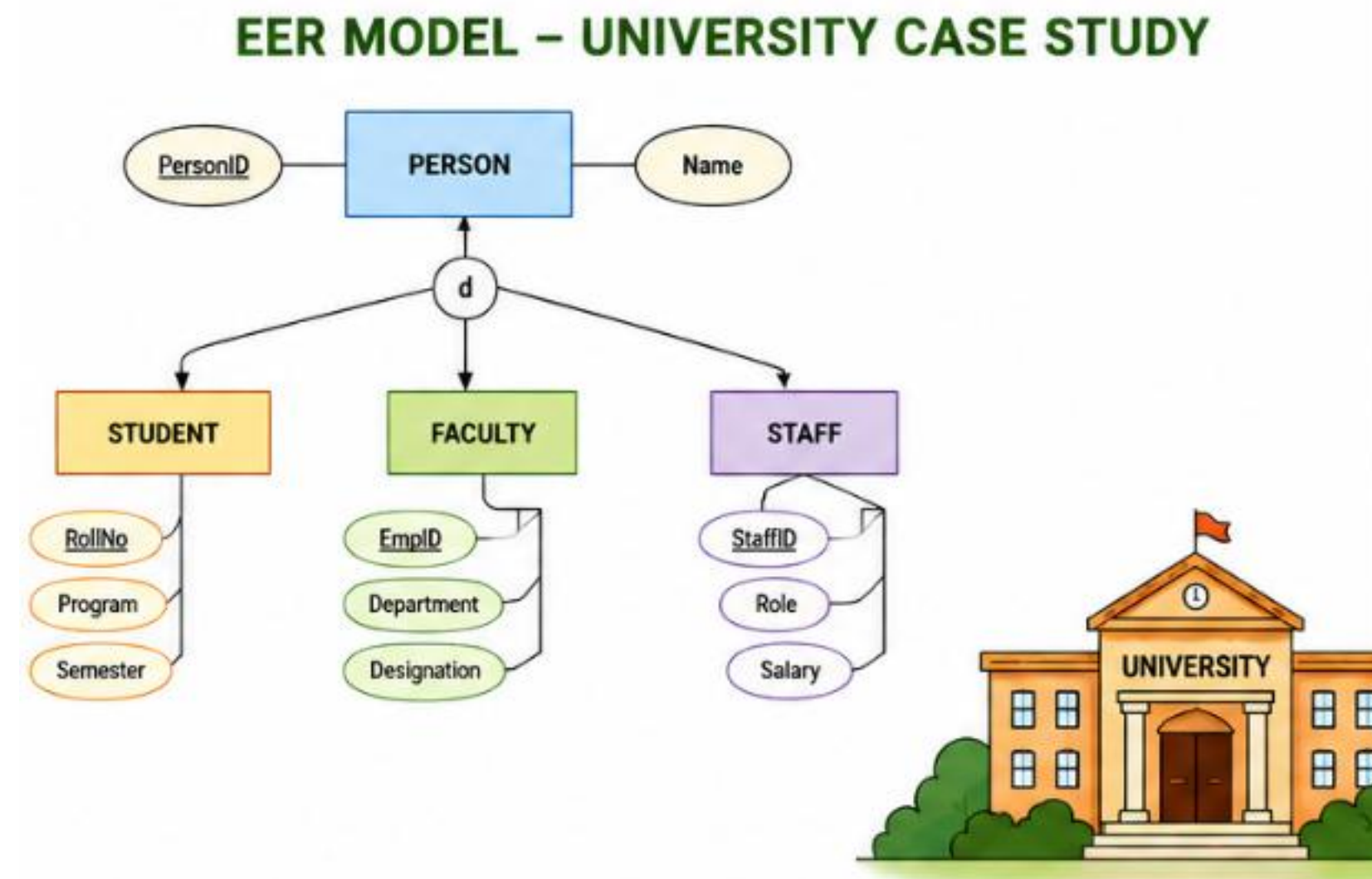
- Student
- Faculty
- Staff

Generalization

- Combines Student, Faculty, and Staff into Person.

Specialization

- Person is divided into Student, Faculty, and Staff.



Advantages & Conclusion

Advantages:

- Represents real-world scenarios.
- Supports inheritance.
- Improves database flexibility.
- Reduces redundancy.

Conclusion:

EER models make database design more effective.



Question 3: Data Independence in DBMS



Introduction

- **Data independence** allows changes to the database without affecting application programs.
- It makes the database **flexible** and easy to update.
- It **reduces maintenance effort** and saves time.
- It **improves performance** by enabling efficient data management.

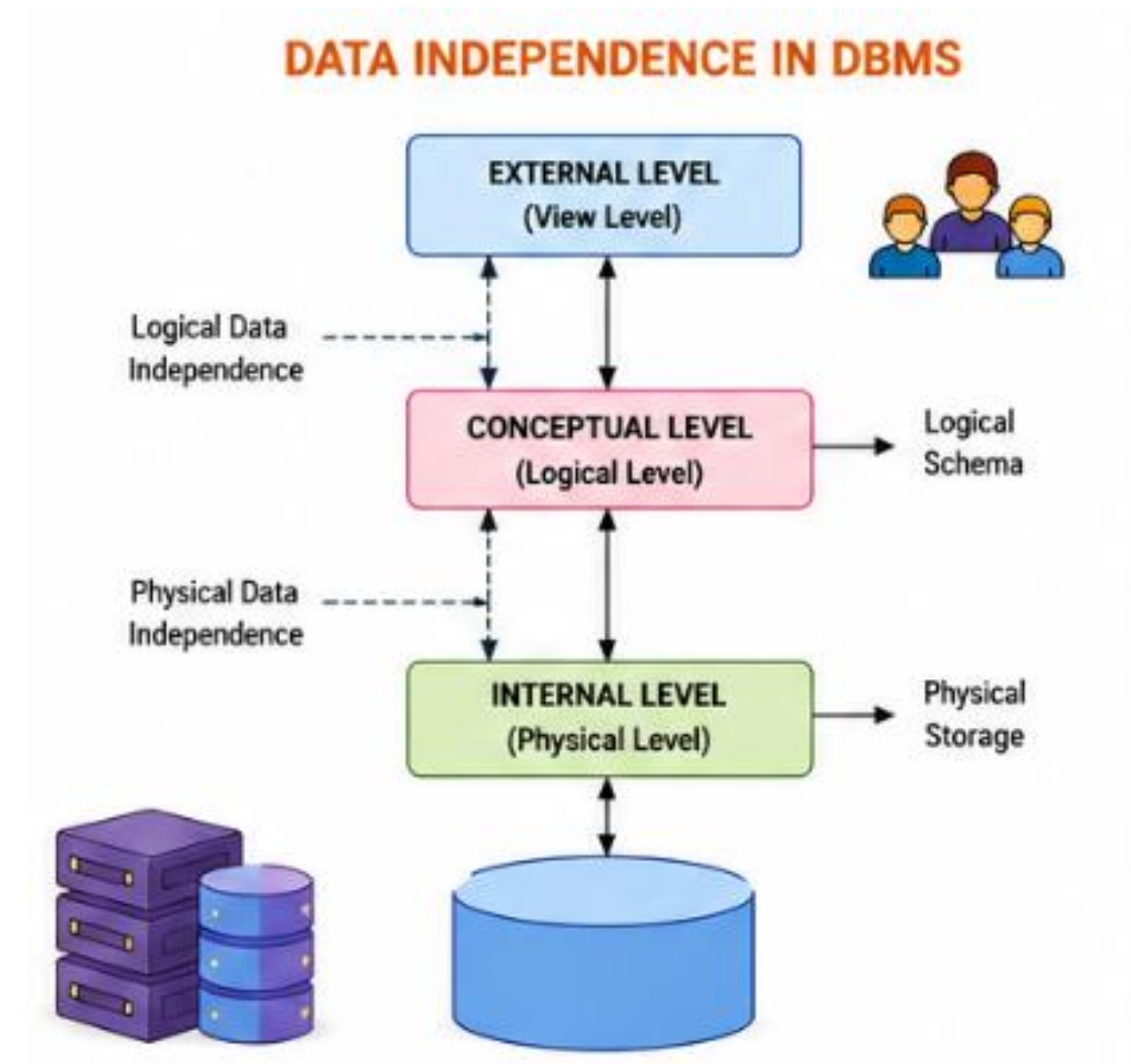
Types of Data Independence

Logical Data Independence

- Changes logical schema.
- No changes to applications.

Physical Data Independence

- Changes physical storage.
- Logical schema remains unchanged.



Advantages & Conclusion

Advantages:

- Easy to modify the database structure.
- Reduces application maintenance.
- Improves database flexibility and scalability.
- Enhances overall system performance.

Conclusion:

Data independence enhances database efficiency by allowing changes without affecting application programs.



Question 4: Software Components of DBMS



Query Processor

Functions:

Parses SQL Queries: Checks the syntax and converts SQL statements into an internal format.

Optimizes Queries: Selects the most efficient execution plan to improve performance.

Executes Queries: Processes the optimized query and performs the requested operations.

Retrieves Required Data: Fetches the requested data from the database and returns the results to the user.

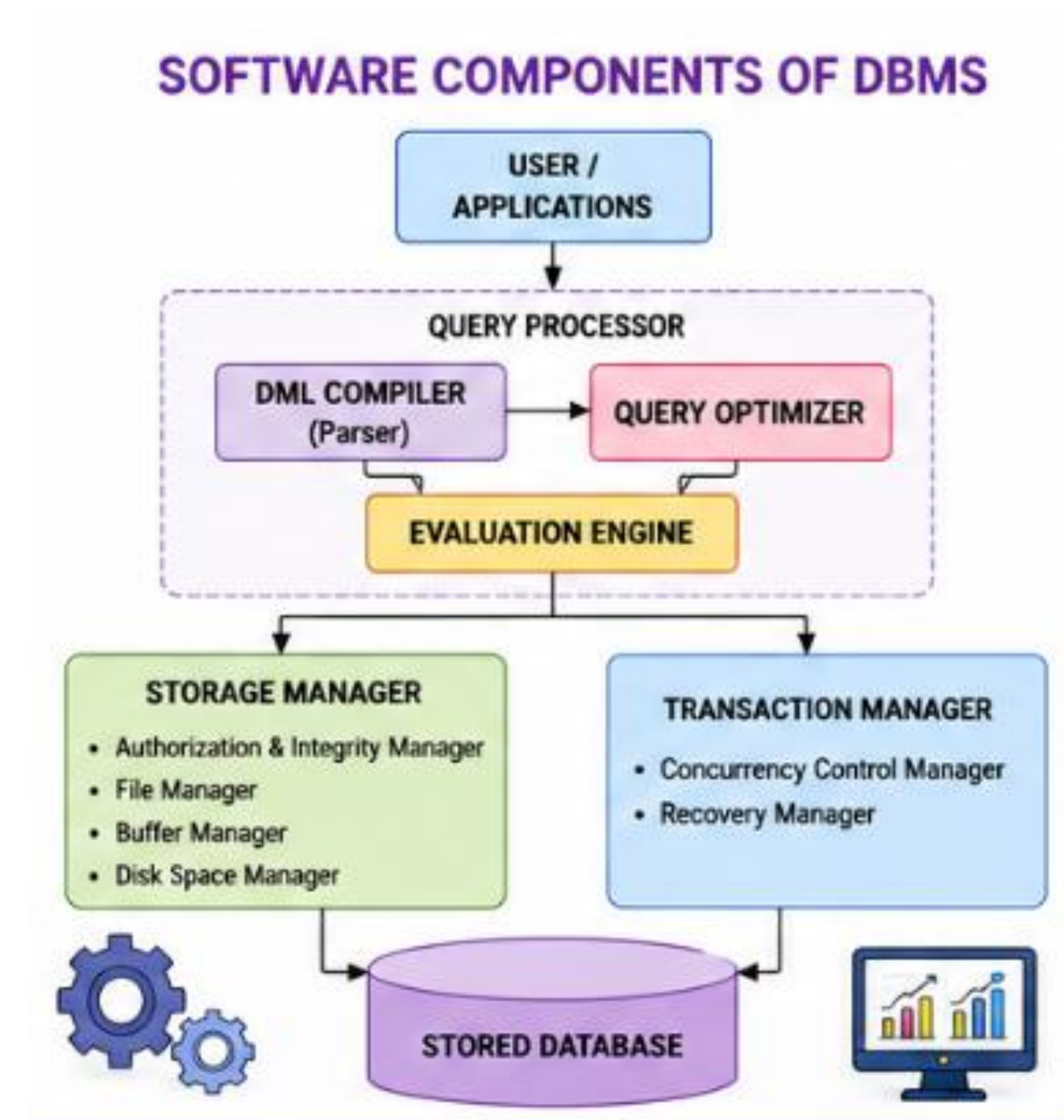
Storage Manager & Transaction Manager

Storage Manager

- Stores data.
- Manages files.
- Handles memory.

Transaction Manager

- Maintains ACID properties.
- Controls concurrent transactions.
- Performs recovery.





Conclusion

- The **Query Processor** interprets and executes SQL queries efficiently.
- The **Storage Manager** organizes, stores, and retrieves data securely.
- The **Transaction Manager** ensures data consistency and maintains ACID properties.
- **Together, these components ensure efficient, secure, and reliable database management.**



Question 5: Strong and Weak Entities



Strong Entity

Features

- Has a **Primary Key** to uniquely identify each record.
- Exists **independently** without relying on another entity.
- Can be **uniquely identified** in the database.

Examples

Student – Student ID

Employee – Employee ID

Customer – Customer ID



Weak Entity

Features

- **Has no Primary Key** and cannot be identified on its own.
- **Depends on a Strong Entity** for its existence.
- **Uses a Partial Key** along with the strong entity's key for identification.

Examples

Dependent – Depends on an Employee.

Order Item – Depends on an Order.

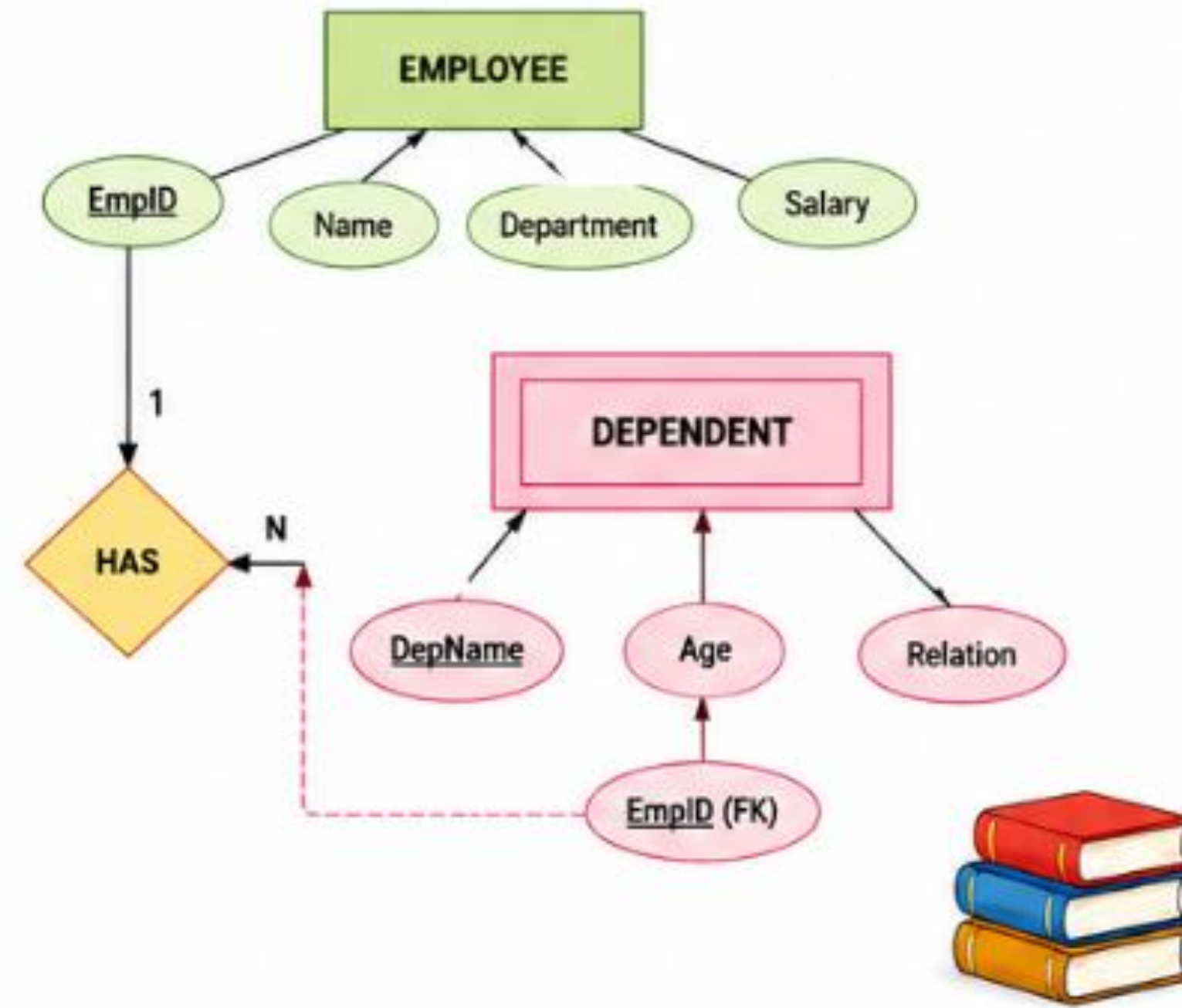
Book Copy – Depends on a Book.



Engineer to Excel



STRONG AND WEAK ENTITIES IN ER MODEL



Comparison & Conclusion

Strong Entity	Weak Entity
Has Primary Key	No Primary Key
Independent	Dependent
Own Existence	Depends on Strong Entity
Single Rectangle	Double Rectangle

Conclusion:

Strong and weak entities help model real-world relationships accurately and improve database design.

Thank
you
Love

